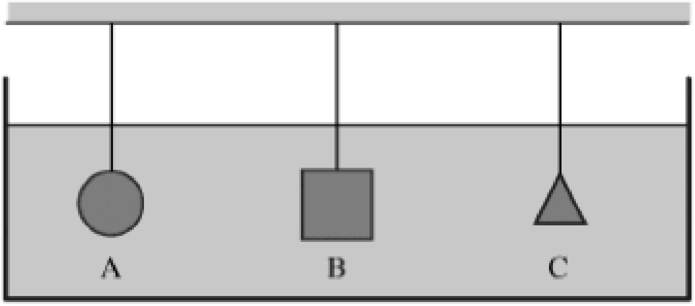
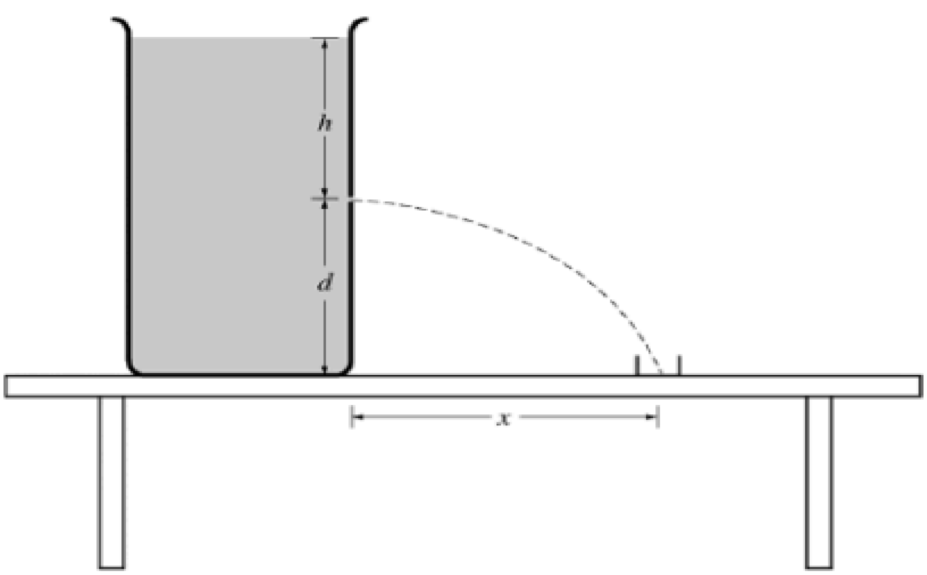
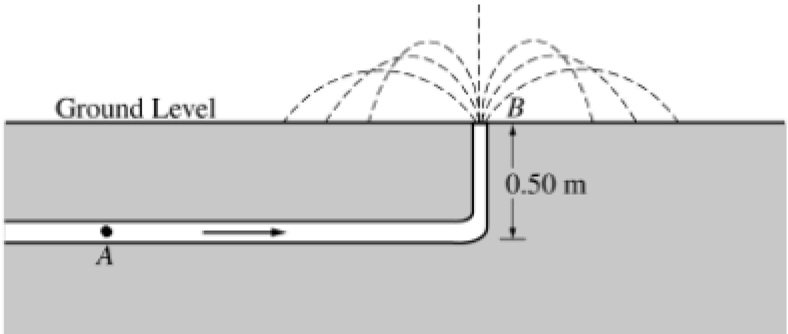
FLUIDS PRACTICE FRQs



Three objects of identical mass attached to strings are suspended in a large tank of liquid, as shown above.
(a) Must all three strings have the same tension?
____ Yes ____ No
Justify your answer.
Object A has a volume of 1.0×10^{-5} m^3 and a density of 1300 kg m^3. The tension in the string to which object A is attached is 0.0098 N.
(b) Calculate the buoyant force on object A.
(c) Calculate the density of the liquid.
(d) Some of the liquid is now drained from the tank until only half of the volume of object A is submerged. Would the tension in the string to which object A is attached increase, decrease, or remain the same?
____ Increase ____ Decrease ____ Remain the same
Justify your answer.

The large container shown in the cross section is filled with a liquid of density 1.1×10^3 kg/m^3. A small hole of area 2.5×10^{-6} m^2 is opened in the side of the container a distance h below the liquid surface, which allows a stream of liquid to flow through the hole and into a beaker placed to the right of the container. At the same time, liquid is also added to the container at an appropriate rate so that h remains constant. The amount of liquid collected in the beaker in 2.0 minutes is 7.2×10^{-4} m^3.
(a) Calculate the volume rate of flow of liquid from the hole in m3 s.
(b) Calculate the speed of the liquid as it exits from the hole.
(c) Calculate the height h of liquid needed above the hole to cause the speed you determined in part (b).
(d) Suppose that there is now less liquid in the container so that the height h is reduced to $h/2$. In relation to the collection beaker, where will the liquid hit the tabletop?
____ Left of the beaker ____ In the beaker ____ Right of the beaker
Justify your answer.





An underground pipe carries water of density 1000 kg/m^3 to a fountain at ground level, as shown above. At point A, 0.50 m below ground level, the pipe has a cross-sectional area of 1.0×10^{-4} m^2. At ground level, the pipe has a cross-sectional area of 0.50×10^{-4} m^2. The water leaves the pipe at point B at a speed of 8.2 m/s.

(a) Calculate the speed of the water in the pipe at point A.

(b) Calculate the absolute water pressure in the pipe at point A.

(c) Calculate the maximum height above the ground that the water reaches upon leaving the pipe vertically at ground level, assuming air resistance is negligible.

(d) Calculate the horizontal distance from the pipe that is reached by water exiting the pipe at 60° from the level ground, assuming air resistance is negligible.